

Checkpoint

12.3 The Dot Product

1 Problems

Problem 1. Find $\langle 2, 6, -1 \rangle \cdot \langle -3, 0, 4 \rangle$.

Problem 2. If $\vec{u} = -2\hat{i} - \hat{j} + \hat{k}$ and $\vec{v} = 5\hat{i} + 2\hat{j} - 2\hat{k}$, what is $\vec{u} \cdot \vec{v}$?

Problem 3. If $\vec{u} \cdot \vec{v} = 2$, $|\vec{u}| = 8$, and $|\vec{v}| = \frac{1}{2}$, what is the angle between $\vec{u}$ and $\vec{v}$?

Problem 4. Are $\langle 1, 1, 1 \rangle$ and $\langle -2, 3, -1 \rangle$ orthogonal?

Problem 5. Find $\text{proj}_a b$ if $a = \langle 1, 0, 2 \rangle$ and $b = \langle 0, 0, 3 \rangle$.

2 Answers

1. -10
2. -14
3. $2 = (8)(\frac{1}{2}) \cos(\theta)$ so $\cos(\theta) = \frac{1}{2} \implies \theta = \frac{\pi}{3}$.
4. Yes since $\langle 1, 1, 1 \rangle \cdot \langle -2, 3, -1 \rangle = 0$
5. $\text{proj}_a b = \langle \frac{6}{5}, 0, \frac{12}{5} \rangle$